TOWSON UNIVERSITY
COLLEGE OF GRADUATE STUDIES

AI-ASSISTED ADOPTION OF NIX FLAKES FOR REPRODUCIBLE DEVELOPMENT ENVIRONMENTS IN COMPUTER SCIENCE EDUCATION

by

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Abstract

AI-ASSISTED ADOPTION OF NIX FLAKES FOR REPRODUCIBLE DEVELOPMENT ENVIRONMENTS IN COMPUTER SCIENCE EDUCATION

Luis Dale Gascon

Inconsistent development environment can slow down software developers and their team. A new hire spends days configuring their systems. Engineers are tasked in resolving environment issues instead of pushing features.
In an educational environment, having students configure their machines for each course can act as a significant blocker to actual learning. An entire lecture period is often sacrificed to the manual and imperative installation of software and dependencies. Personal devices introduce unknown variables such as their operating systems, configuration, or hardware, resulting in students having to be troubleshoot problems specific to their devices that puts an instructor as a role of tech support for devices they might not have experience with.

I propose a solution that leverages agentic LLMs to synthesize a nix environment via nix flakes. We first provide an instructor's natural language description, then the agent queries an mcp-nixos server to discover valid package names and configuration options from real time data. The output is validated via nix evaluation commands, where the functional nature of nix provides a deterministic success or failure signal. In the event of a failure, the LLM uses the error trace as context to correct itself until a valid, reproducible environment is produced.

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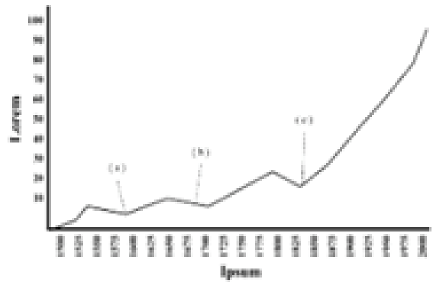

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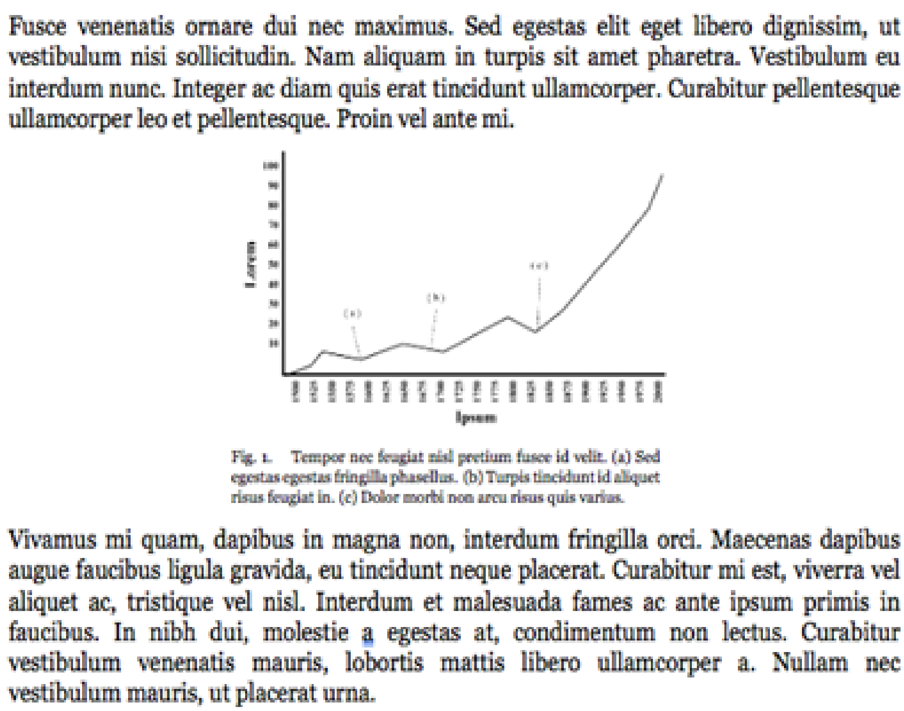
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Appendix A

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Appendix B

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References

[1] W. Du, J.-M. Gratien, R. Gayno, and B. Raffin. Nix to the Rescue for a Reproducible HPC-AI Software Stack. [Online]. Available: http://arxiv.org/abs/2607.26688

[2] S. Valstar, W. G. Griswold, and L. Porter, "Using DevContainers to Standardize Student Development Environments: An Experience Report," in Proceedings of the 2020 ACM Conference on Innovation and Technology in Computer Science Education. ACM, pp. 377–383. [Online]. Available: https://dl.acm.org/doi/10.1145/3341525.3387424

[3] K. G. Srivatsa, S. Mukhopadhyay, G. Katrapati, and M. Shrivastava, "A Survey of using Large Language Models for Generating Infrastructure as Code."